MomCheck

Voice follow-up for every pregnant woman and new mother, in her own language, with her ASHA worker in the loop.

Live app momcheck.onrender.com
Code github.com/Shubhz20/momcheck
Open a demo ASHA account. No sign-up needed.

THE PROBLEM

Most maternal deaths in India come from **delays**: in recognising a danger sign, in reaching care, and in getting care. The person meant to catch the first delay is the **ASHA worker**. She looks after every pregnant woman and newborn in her area, mostly on foot, and often in a language that isn't the mother's own, since urban wards have migrant families from all over India. She cannot ask every woman about every danger sign every week.

WHAT MOMCHECK DOES

- **Follow-up conversations in 10 Indian languages**, code-mixed speech included. MomCheck goes through the government's danger-sign checklist (MCP card, Home-Based Newborn Care after birth, and a gentle mood check) one question at a time.
- **Red / Amber triage with an action card.** Every danger sign is graded. For a red sign she is told to go to *her* named hospital now (108/102 free ambulance). The ASHA is alerted **during the conversation** with the next steps: call her, the hospital, and her ANM. A live checklist and risk meter fill in as she talks.
- **In the ASHA's own voice.** The ASHA reads a consent passage once, and **Gnani voice cloning** then speaks to her women in her voice, in any of the 10 languages. Women answer a voice they trust, and MomCheck always says it is an automated assistant.
- **Interpreter for home visits.** The ASHA speaks Marathi, and the Tamil mother hears Tamil in the ASHA's voice. Her reply comes back translated, and danger signs still raise alerts.
- **Ask MomCheck.** Each mother gets a personal link or QR code to ask questions in her language whenever she is worried. It needs no app or login, and her ASHA hears about any danger sign.
- **For the ASHA:** spoken visit notes become records, she gets a spoken daily briefing, and she has a risk map of her women and a 7-day impact panel.

BUILT ON GNANI AI

Her voice → Prisma STT → Gemini → Timbre / cloned voice

GNANI MODEL	HOW MOMCHECK USES IT
Prisma STT `prisma-v3.0`	Every turn, voice notes, interpreter, consent check. Her language plus English for code-mixed speech, with key terms (her name, village, PHC).
Timbre TTS `timbre-v3.0`	MomCheck's voice in 10 languages, the daily briefing, and 8 kHz audio for phone lines.
Voice cloning `vachana-vc-v1`	The ASHA's voice from one sample, speaking every language.

Gemini does the reasoning (next question, grading, translation, summaries). The conversation loop is our own.

EVIDENCE: NOISY PHONE LINES

30 synthetic replies in 10 languages were spoken by Timbre, squeezed to an 8 kHz μ-law phone line with street noise, and transcribed by Prisma.

CONDITION	CHAR. ERROR	DANGER SIGN CAUGHT
Clean, 16 kHz	6.8%	**30/30**
Phone line, 8 kHz	7.3%	**30/30**
+ street noise, 10 dB	7.3%	**30/30**
+ street noise, 5 dB	11.4%	**30/30**
+ noise as loud as her voice, 0 dB	28.3%	**25/30**

Prisma writes Hinglish naturally: “मुझे थोड़ा bleeding हो रहा है”.

SAFETY AND RESPONSIBILITY

MomCheck screens and escalates. It never diagnoses or prescribes, and every danger sign reaches a human health worker. The protocol follows MCP and HBNC guidance. Voice cloning needs the ASHA's read-aloud consent, which Prisma verifies, and she can delete her voice at any time. On a shared phone MomCheck reveals nothing about her health. All demo data is synthetic.

STATUS AND NEXT STEPS

Live today on free infrastructure, with two demo ASHA accounts (rural Hindi, and an urban ward speaking nine languages). Real phone calls (Twilio Media Streams, 8 kHz μ-law) are built and tested against a simulated stream. **Next:** Gnani streaming APIs to cut the roughly 4 s reply time, a pilot with real ASHAs, and an ANM supervisor view.